ALEXIS at DialectSentEval Subtask 2: Retrieval-Augmented, Preference-Tuned Generation for Arabic Sentiment Swap

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Abstract

Sentiment swap is the task of rewriting a sentence so that it carries the opposite polarity while its topic stays the same. We present a system for Arabic sentiment swap built on two observations about the data. First, the swap is a local edit rather than a paraphrase: 91% of reference rewrites stay within two words of their source. Second, the polarity to swap *away from* has to be predicted, because the test data does not provide it. We therefore fine-tune mT5 (Xue et al., 2021) and refine it with Direct Preference Optimization (Rafailov et al., 2023) under a reward that balances target-polarity probability against content preservation. At inference we pool candidates from four generators, one of them a minimal-edit tagger (Malmi et al., 2019; Mallinson et al., 2020), together with a retrieval index, rerank them with the same balance, and repair them with polarity-flip lexicons mined from the training set. Source polarity is supplied by a dedicated classifier that reaches 97.3% accuracy, against 74.3% for the off-the-shelf sentiment model that the evaluation itself uses. Every component was scored on the official leaderboard: generation-side work contributes +0.044 Sentiment Style Accuracy, and the dedicated polarity classifier a further +0.155. The final system reaches 0.9581 Sentiment Style Accuracy, 56.91 BLEU and 72.18 chrF, ranking **1st** of 15 submissions.

1 Introduction

Sentiment swap asks for a rewrite that inverts a sentence's polarity and changes as little else as possible. Two objectives pull against each other: the output must land firmly on the opposite side of the polarity boundary, yet stay close to the source, since content preservation is measured against a human rewrite that is itself a small edit. Push too hard on polarity and the rewrite drifts off topic; push too hard on similarity and the polarity does not flip. Most work on this problem is in English; Arabic has received far less attention (Section 2).

Subtask 2 of DialectSentEval (Ezzini et al., 2026a), held with the Arabic Natural Language Processing Conference (ArabicNLP 2026),[1] provides a benchmark for this setting and continues the sentiment-swap component piloted in AraSentEval (Ezzini et al., 2026b). It uses the MA'AKS dataset (Mughaus et al., 2026): 9,647 Modern Standard Arabic sentence pairs, split 6,752/964/1,931 into train, validation and test, whose members share a topic but carry opposite polarities. Systems are scored on Sentiment Style Accuracy (SSA), the fraction of outputs whose polarity matches the target according to a fixed CAMeLBERT sentiment classifier (Inoue et al., 2021), together with BLEU (Papineni et al., 2002) and chrF (Popović, 2015) against the reference rewrite.

One detail of the released data shaped the whole system. The task description says the source sentence's polarity is given as input, and the train and validation files do contain a `source_polarity` column, but the test file does not. Source polarity therefore has to be predicted at inference time. The obvious way to do that is to reuse the classifier that grades the task, and that is what we did at first. Measured against gold labels, it gets the source polarity right only 74.3% of the time. Roughly a quarter of the test set was being rewritten towards the wrong target, and no amount of tuning on the generator could see the problem.

This paper makes four contributions:

- A minimal-edit treatment of Arabic sentiment swap, motivated by corpus statistics and realized through retrieval, an edit-tagging model, and a content term in the reranker.
- A dedicated source-polarity classifier for the label the test set omits: 97.3% against 74.3%

[1] https://ezzini.github.io/DialectSentEval2026/. Our code: GitHub.

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for the evaluation classifier, worth +0.155 SSA.

- A DPO reward combining polarity confidence with content preservation, and evidence that it must use the raw three-way probability rather than a renormalized binary one.
- A cumulative ablation in which every configuration was scored on the official leaderboard, not a local proxy.

2 Related Work

Text style transfer. Sentiment transfer is usually studied without parallel data, so methods either disentangle content from attribute by adversarial cross-alignment (Shen et al., 2017), or delete attribute markers, retrieve replacements and regenerate (Li et al., 2018); Jin et al. (2022) survey the field. MA'AKS is unusual in supplying human-written parallel rewrites. We can therefore fine-tune a sequence-to-sequence model directly, and keep the marker-lexicon idea of Li et al. (2018) for retrieval and post-processing rather than making it the main model. Where source and target overlap heavily, token-level tagging models (Malmi et al., 2019; Mallinson et al., 2020) are also a natural fit; we use one as a pool member. Rewriting under a fixed topic is also central to Arabic conversational question rewriting (Hassib et al., 2025), though there the rewrite target is context completion rather than sentiment.

Evaluation. Style accuracy from a fixed classifier is the standard metric, but classifier-based evaluation is fragile, and optimizing hard against an imperfect proxy degrades true performance (Gao et al., 2023). Sections 4 and 6 report a concrete instance of each.

Sentiment analysis. Arabic sentiment analysis is well served by pre-trained encoders (Inoue et al., 2021), and Arabic-specific generative models exist (Nagoudi et al., 2022), but Arabic style transfer remains comparatively unexplored.

3 Task and Data

MA'AKS (Mughaus et al., 2026) pairs each source sentence with a human-written rewrite of the opposite polarity (Table 1). Sources lean slightly negative (3,731 against 3,021 in training), and the register is noisy social media text, with emoji in 60.7–63.7% of sentences. Three properties of the corpus each motivated a component of our system.

	Train	Val	Test
Pairs	6,752	964	1,931
Unique sources	5,517	–	–
Has `source_polarity`	Yes	Yes	**No**
Exact dup. of train source	–	20.4%	22.8%

Table 1: Dataset splits. Unlike train and validation, the test file carries no `source_polarity` column.

Repetition. Sources repeat across splits: 22.8% of test sentences are byte-identical to a train or validation source. For those, a gold rewrite already exists, so retrieval supplies a strong candidate at no modelling cost.

Locality. The swap is a local edit. 91.0% of training pairs are within two words of their source's length, and character-level similarity between source and target is tightly concentrated (median 0.82, interquartile range 0.75–0.88). A training pair shows the pattern: تجربة جميلة سعر كتير مناسب شكرا لكم (*'a nice experience, reasonable price, thank you'*) becomes تجربة سيئة سعر كتير غير مناسب لا شكرا علكم (*'a bad experience, unreasonable price, no thanks'*). A few words change; the structure and length do not. Constrained editing is therefore closer to what the references look like than open-ended paraphrasing.

Lexical regularity. The same polarity markers flip over and over. We mined 323 aligned emoji flips (broken-heart → sparkling-heart and similar) and short antonym pairs such as كريهة → جميلة ('hideous' → 'beautiful', 115 occurrences). This gives a small deterministic lexicon that can catch markers a neural model leaves behind.

4 System

The pipeline has three stages (Figure 1). We predict the polarity of the source, which fixes the target polarity t that everything downstream aims at; we collect candidate rewrites from four generators and a retrieval index; and we score the pool, return the best candidate, and repair what the models left behind.

One scoring function ties the last two stages together. For a source x and a candidate rewrite y,

$$s(y) = \alpha\, p_t(y) + (1 - \alpha)\, \mathrm{chrF}(y, x), \quad (1)$$

where $p_t(y)$ is the probability the evaluation classifier assigns to the target polarity t, read off its

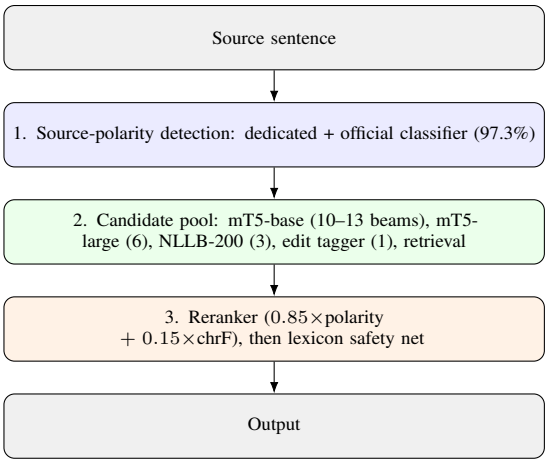


Figure 1: Inference pipeline. Retrieved near-duplicates are ordinary candidates, not a bypass, and four generators feed a single reranker.

raw three-way softmax over positive, negative and neutral. The two terms are the competing objectives of Section 1, and α sets the trade-off; we use $\alpha = 0.85$ when reranking and the same form as the DPO reward. Putting the grading classifier inside s is deliberate: from here on, agreeing with the grader is the objective.

Source-polarity detection. The test file omits `source_polarity`, so we predict it and set t to the opposite label. Using the grading classifier zero-shot is the self-consistent choice, but on this job it is right only 74.3% of the time. A classifier fine-tuned on the 7,716 labeled pairs reaches 96.7%, and since the two fail in different places, a 50/50 vote between them reaches 97.3%.

Primary generator. We fully fine-tune `google/mt5-base` (Xue et al., 2021) on all 7,716 pairs and decode with 10 beams, or 13 when the source is predicted negative. Negative-to-positive is the harder direction on validation (chrF 78.94, SSA 94.37%, against 84.27 and 96.06% the other way), an asymmetry also reported for sentiment transfer more broadly (Mukherjee et al., 2022). We select checkpoints by the chrF of beam-search output rather than by teacher-forced loss, because the two disagreed in an early run: the lowest-loss checkpoint wrote measurably worse text once it had to decode freely. Two Arabic-specific T5 checkpoints (Nagoudi et al., 2022) were tried and discarded: an unparseable tokenizer format in one, incoherent zero-shot output in the other.

Preference optimization. We refine the generator with Direct Preference Optimization (Rafailov et al., 2023) on 4,500 examples. For each we sample four candidates, add the gold rewrite, score all five with Equation 1, and train the model to prefer the higher-scoring member of a pair. An early version of the reward dropped the neutral class and renormalized the remaining two probabilities. That version is exploitable: a rewrite can look confidently on-target once neutral is ignored while the grader still reads it as neutral, so the model learns to satisfy the proxy instead of the objective (Gao et al., 2023). Keeping the raw three-way probability closes the loophole, and Section 6 measures what it was costing.

Widening the pool. Three further systems contribute candidates, each failing differently from mT5-base. `facebook/nllb-200-distilled-600M` (NLLB Team et al., 2022), fine-tuned as a monolingual Arabic-to-Arabic "translator", adds architectural diversity (3 beams). An `mT5-large` instance of the same recipe adds capacity (6 beams). An edit-tagging model in the style of LaserTagger (Malmi et al., 2019) and FELIX (Mallinson et al., 2020) labels each source token KEEP, DELETE or REPLACE-with-a-mined-word; it cannot paraphrase at all, which is why it helps, and still reaches the reference for 90.6% of validation sources. Retrieval, in the spirit of Li et al. (2018) and of sentence-embedding similarity scoring more generally (Ibrahim et al., 2023), contributes the gold rewrite of any training source within 0.90 cosine similarity of x, covering the 22.8% of test sentences that recur verbatim.

Reranking and repair. Every candidate, retrieved ones included, is ranked by Equation 1 and the top one is returned; a retrieved rewrite has to win on the same terms as a generated one. A final deterministic pass applies the mined lexicons of Section 3, flipping any emoji or antonym that still carries the source polarity.

5 Experimental Setup

Table 2 lists the deployed configuration. The submitted generator is trained on all 7,716 train and validation pairs; validation figures quoted during development come from the held-out 964-row split. We grid-searched α and the near-duplicate threshold on validation, but that signal was misleading on one axis. Its best setting was $\alpha = 1$, all weight on polarity and none on content, and submitting it lowered BLEU and chrF on the leaderboard. We

therefore moved part of the way towards each validation optimum rather than adopting it outright.

Hyperparameter	Value
Primary generator	mT5-base
Primary beams (std. / hard dir.)	10 / 13
NLLB-200 beams (pooled)	3
mT5-large beams (pooled)	6
DPO preference examples	4,500
DPO candidates/example	4 sampled + gold
DPO epochs	7
Reranking weight α (polarity)	0.85
Reranking weight $1-\alpha$ (chrF)	0.15
Retrieval similarity threshold	$\geq$0.90 (cosine)

Table 2: Key hyperparameters of the final deployed system.

6 Results

Our submission reached SSA = 0.9581, BLEU = 56.91 and chrF = 72.18, first of fifteen submissions and 17.3 SSA points ahead of the second-placed system (Table 3). The margin is specific to SSA: several systems are within a few points on BLEU and chrF and differ mainly in whether the polarity actually flipped.

#	Team/Participant	SSA	BLEU	chrF
1	**ALEXIS (momen_ibrahim)**	**0.9581**	**56.91**	**72.18**
2	akarrouch	0.7851	49.82	70.12
3	ahmadalmoustafa	0.7789	41.48	64.45
4	mona_	0.7597	51.58	71.02
5	melmoussaoui	0.7530	49.03	69.53
6	mohamedsabaa	0.7447	49.07	69.33
7	hsnhb7	0.7343	46.80	67.10
8	areejkhaled	0.6898	54.63	72.61
9	mohammed_bahgat	0.6732	41.38	65.84
10	ehabashraf	0.6732	41.38	65.84
11	ahmad_alshattha	0.6354	25.07	56.33
12	hoanguitcs	0.6220	49.27	68.26
13	normalman159	0.5997	11.31	38.22
14	mustafaelkady	0.4772	0.03	10.04
15	mohamedmhaouach	0.2807	43.06	64.61

Table 3: Official Subtask 2 leaderboard. Our submission is shown in bold.

What each component contributed. Table 4 traces the system's development. Every row was submitted and scored on the leaderboard, so the gains are real test-set gains rather than validation estimates. Starting from mT5-base with retrieval and reranking (0.7452), the edit tagger and a wider candidate pool added +0.017, and DPO together with the NLLB-200 and mT5-large generators added a further +0.027. Replacing the evaluation classifier with our dedicated one for source-polarity detection added +0.155, more than three times the two generation-side gains combined. Correcting p_t, first in the reranker and then in the DPO reward, added the last +0.015.

The reason is simple: if the source polarity is wrong, the system rewrites a negative sentence to be more negative, and no beam width or reranking weight recovers from that. Everything before this fix was tuning how well the system reached a target that was wrong on a quarter of the test set.

Configuration	SSA	Δ
mT5-base + retrieval + reranking	0.7452	–
+ edit tagger, wider pool	0.7618	+0.017
+ DPO, NLLB & mT5-large	0.7887	+0.027
+ **dedicated polarity clf.**	**0.9436**	**+0.155**
+ hard polarity gate	0.8933	−0.050
+ raw-softmax reranking	0.9498	+0.006
+ DPO reward fix (final)	**0.9581**	+0.008

Table 4: Cumulative ablation; every row is an independent leaderboard submission. Δ is measured against the previous *retained* configuration: the hard polarity gate was reverted after it regressed, so the two following rows are measured from 0.9436.

A change that looked safe and was not. One row in Table 4 is a regression. We added a hard gate that discarded any candidate the classifier did not label with the target polarity. It appeared to cost nothing on held-out validation, so we submitted it, and the score fell by 0.050. The flaw was in the check: the same classifier chose the candidates and judged them, so it could not detect where it disagrees with the grader. A reranking rule validated only against the signal it optimizes will always look correct. We reverted the gate and moved to the raw three-way softmax, which recovered the loss and then some.

Cost. Running four generators, a tagger and a retrieval index at inference is expensive, and Table 4 shows where to economise: the pooled generators are worth about +0.044 in total, against +0.155 for the polarity classifier alone. A latency-sensitive deployment could keep only mT5-base and the classifier.

7 Conclusion

We presented a system for Arabic sentiment swap that treats the task as constrained editing and makes the target polarity explicit. One scoring function holds polarity and content in balance, serving both as the DPO reward and as the reranker over a pool

of four generators and a retrieval index. The system ranked first of fifteen submissions.

The largest single gain, however, came from predicting a label the task assumed would be given. The natural substitute, the classifier that grades the task, is unreliable at that job. When a shared task is built around an auxiliary classifier, it is worth measuring that classifier on the sub-task it is asked to perform before trusting it.

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